

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0462

Goodhue Solar 100 kV
May 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0462. The Interconnection Customer proposes a 230.4 MW natural gas combined cycle facility interconnecting at the Goodhue Solar 100 kV substation in the Indiana area. The request was received on April 17, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,882,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ligonier to Goodhue Solar 100kV	Brunner Island to Co Line Road 500kV	209.1	122	100.0
Cuba City to Farmersville 100kV	Cypress Crk to Ligonier 100kV	133.3	129	100.0
Ligonier to Cypress Creek 100kV	Ligonier to Montibello 100kV line	100.8	232	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ligonier to Goodhue Solar 100kV	Rebuild line with higher-capacity conductor	\$4,191,000	100% (\$4,191,000)
Cuba City to Farmersville 100kV line	Rebuild line with higher-capacity conductor	\$8,133,000	100% (\$8,133,000)
Cypress Crk to Ligonier 100kV	Reconductor line	\$2,558,000	100% (\$2,558,000)
	Total allocated to project		\$14,882,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.